

**An AI-Powered Virtual Assistant for Personalized Clothing
Selection Using 3D Body Modeling**

25-26J-302

Project Proposal Report

Shangeetha J

B.Sc. (Hons) Degree in Information Technology Specializing in
Information Technology

Department of Information Technology

Sri Lanka

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DECLARATION

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ABSTRACT

The growth of e-commerce has transformed the way consumers engage with fashion, but current virtual try-on technology is not designed to offer an inclusive, personalized, and emotionally responsive dressing experience. Current solutions are mannequin-based and rooted in static installations, do not provide realistic skin tone and body representation, and are not sensitive to differently abled users' accessibility needs. These limitations reduce realism, inclusivity, and consumer confidence and contribute to high return rates in e-commerce shopping.

The proposed system, Style Mate: An AI-Powered Virtual Assistant for Personalized Clothing Selection, addresses these challenges by integrating computer vision, machine learning, natural language processing (NLP), and affective computing into one unified architecture. The system offers customized 3D mannequins generated from simple 2D images, accurate skin tone analysis, emotion-aware outfit recommendations, and a chatbot to increase accessibility.

Style Mate's aim is to democratize fashion technology to enable users of all body shapes, cultures, and abilities to make decisions on style in an informed and confident manner. The desired outcomes are improved virtual try-on accuracy, increased inclusivity, and customer confidence, and decreased retailers' product return rates and improved research in AI-powered fashion and human-centered system design.

KeyWords: Virtual Try-On, Personalized Fashion, 3D Modeling, Emotion-Aware Recommendation, Inclusive

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LIST OF ABBREVIATIONS

Abbreviation	Description
VTO	Virtual Try-On
AI	Artificial Intelligence
HSV	Hue, Saturation, Value
ML	Machine Learning
WCAG	Web Content Accessibility Guidelines
NLP	Natural Language Processing
SRS	Software Requirements Specification
2D	Two-Dimensional
IDE	Integrated Development Environments
3D	Three-Dimensional
GAN	Generative Adversarial Network

1 INTRODUCTION

1.1 Background and Literature Survey

The fashion industry has rapidly shifted to digital media with the introduction of e-commerce, fundamentally transforming how individuals interact with apparel and accessories. In this context, Virtual Try-On (VTO) systems have emerged as promising alternatives for enabling customers to preview clothing items without physically trying them on. The ultimate goal of these technologies is to increase consumer confidence, reduce product return rates, and enhance overall satisfaction in online shopping [1].

However, despite significant progress, existing virtual dressing technologies typically do not provide truly personalized, realistic, and inclusive experiences. Most VTO systems use static mannequins or generic avatars that do not capture the diversity of human body shapes, poses, and cultural backgrounds. It has been suggested that this simplification leads to lower trust in virtual fashion technologies, particularly for customers who have non-normative bodies or mobility impairments [2]. Moreover, the majority of the systems are not responsive to skin tone and texture, resulting in unrealistic visualizations that cannot appeal to diverse groups of users. Not only does this limitation affect user satisfaction, but it also hinders the inclusivity of fashion technology [3].

Current research identifies growing interest in utilizing deep learning and computer vision for enhancing personalization in virtual fashion. For instance, convolutional neural networks and generative adversarial networks (GANs) have been utilized for clothing transfer and multi-pose outfit visualization with improved realism [4]. Other work has included body shape classification models and 3D reconstruction techniques to create mannequins that better reflect individual user measurements [5]. While these techniques are important steps forward, they remain computationally expensive and beyond the reach of everyday users, as many still require specialized hardware such as 3D scanners.

Beyond physical appearance, the emotional and experiential dimension of digital fashion has also been investigated. Research in affective computing shows that incorporating user emotions, mood and facial expressions into recommender systems

can radically enhance engagement and relevance [6]. Yet, affective fashion recommendation remains an unexplored area since most current systems are solely focused on visual fit and not on users' psychological and contextual interests. Similarly, the accessibility aspect of VTO technology is typically neglected, so individuals with physical or cognitive impairment are not adequately catered to by mass-market products [7].

In summary, while existing research demonstrates the high potential of virtual try-on systems to revolutionize online shopping, most academic and commercial solutions still lack personalization, inclusivity, realism, and emotional awareness. This gap inspires the development of advanced frameworks that organically integrate body-aware modeling, skin tone adaptation, emotion-sensitive recommendations, and accessible interaction modalities into a unified, intelligent fashion assistant. Filling in these gaps not only makes shopping online more fun but also democratizes fashion technology so that it is equitable and enjoyable for all to utilize.

1.2 Research Gap

With faster computer vision and generative model advancements, many virtual try-on (VTO) proofs have been created. Initial 2D overlay and follow-up GAN/pose-transfer pipelines improved visual realism but tend to depend on generic mannequins and idealized poses, which block realism for customers with complicated body types or non-standing poses. Even more recent pipelines boasting strong garment warping still take full-body, front-facing inputs and clean backgrounds, which puts a bound on their applications in reality. As a result, users who are seated, partially visible, or differently abled remain underserved.

These contributions support body modeling through learning-based body shape categorization and parametric avatars. While these efforts report good technical performance on test images, they rarely consider consumer-level inputs (e.g., occluded phone photographs), nor do they aim specifically at non-ideal capture situations (seated/occluded). Hence, reliable shape estimation from simple 2D inputs without scanners or multiple calibrated views is still lacking for real users [8], [9].

Another area explores 3D scanning/photogrammetry to create highly accurate avatars. They are very accurate but hardware-dependent, slow, and costly, so they are not practical for mass e-commerce use and not accessible to most consumers. This leaves the potential for low-resource, lightweight mannequin generation from standard images that also preserves user privacy and ease of use.

Appearance realism is also constrained by handling skin tones. Most systems output clothing onto default/light mannequin textures or ignore skin altogether, losing inclusivity and authenticity. Very few works incorporate robust skin segmentation + tone classification as part of the mannequin pipeline for equal visualization for diverse users, a fundamental usability and trust consideration that remains unresolved in mainstream VTO literature.

Personalization work on fashion suggestion has progressed (style/occasion matching, visual similarity, history-based learning of preferences). However, effective and context

awareness of what the user perceives and what message the user wishes to express are hardly incorporated into the try-on loop. There is little evidence that suggests emotionally adaptive interfaces can increase satisfaction, but that capability is hardly joined with live VTO previews to co-adapt what the user sees and what the system recommends [10], [11].

Finally, accessibility is handled inconsistently. Most pipelines by far assume standing, full-body, two-hand device operation, and unimpaired vision/cognition. Accessible HCI research prescribes principles and patterns, but these are only partially realized in VTO tools (e.g., voice I/O, TTS, minimal-touch flows, seated capture). This leaves an ongoing usability gap for differently abled users that new systems should fill [12].

The problems of geographically specific systems

Legal decision support systems and legal advisory systems built with a focus on a particular country's legal system do not suit another country's legal system. This could be understood by the points illustrated above.

- **Research A (2D/GAN-based VTO)** - Improves garment warping/fit visualization on perfect inputs; problem: based on default body templates and front, full-body images → suffers from partial/seated users; inclusivity is affected [8].
- **Research B (Body-shape classification)** - Like Learns categories and predicts measurements; problem: trained on posed curation; robustness to casual mobile photos (lighting/occlusion) and non-standing users is low [9]

- **Research C (3D scan/photogrammetry VTO)** - Extremely accurate avatars; problem: requires special hardware/multi-view capture; not scalable or cost-effective for typical online shoppers.
- **Research D (Personalized outfit suggestion)** - Captures style/occasion liking; problem: lacks real-time emotion/context feedback tied to try-on preview; suggestions are not affect-adaptive [10].
- **Research E (Emotion-aware UI for fashion)** - Demonstrates benefit of real-time effect; problem: rarely combined with mannequin generation + try-on process to co-adapt visualization and suggestions [11].
- **Research F (Accessible HCI for fashion apps)** - Establishes accessibility features and guidelines; problem: VTO pipelines seldom have seated/partial capture, voice/TTS, or low interaction flows end-to-end enabled [12].

This gives us a clear understanding of the current systems available, and the problems and gaps present in them. Existing VTO pipelines do not have a shared, lightweight, and accessible solution that

- generates personalized mannequins from simple 2D images (sitting/partial),
- preserves skin-tone variety for realistic visualization,
- integrates affect-aware recommendations with live try-on preview, and

Feature	Existing Systems	Proposed System	Gap Addressed
Hardware	High (3D Scanners, Depth Cameras)	Low (Standard RGB Camera)	Accessibility & Cost
Pose Requirement	Rigid, Standardized (T-pose)	Flexible (Seated/Standing)	Inclusivity for - All
Skin Tone Consideration	Limited / Generic	Core Feature with ML Analysis	Diversity & Realism
Primary Focus	Geometric Accuracy	Inclusivity & User Experience	Market Need
Deployment	Complex, Offline Software	Web-Based, API-Driven	Scalability

Table 1: Research gap illustration

2 RESEARCH PROBLEM

The central research problem addressed in this section is the systemic exclusion and shortage brought about by the dominant technological paradigm of virtual try-on (VTO) and body modeling solutions. Rather than a technical shortcoming, this is an issue with numerous aspects that have dire socio-economic, psychological, and ethical consequences. The rampant reliance on specialized hardware or specific physical capability to create a singular digital avatar imposes an imposing digital divide, actively barring a vast worldwide population from the benefits of the technology and solidifying existing disparities.

This is a problem in many significant and interconnected spheres:

2.1 Socio-Technical Exclusion and the Accessibility Crisis:

The most direct breakdown of current systems is that they are designed to be inaccessible. The usual requirement that a user have a specific, full-body pose (e.g., a "T-pose") dressed in tightly fitting clothing is not a hardship but an unpassable hurdle for individuals with mobility disabilities, chronic pain, certain physical disabilities, or advanced age. This design flaw de facto disenfranchises a wide population, excluding them from access to a tool that would significantly enhance their independence, self-expression, and consumer power on the internet. In the era of digital inclusion, the reality that such ableist design principles have endured in cutting-edge technology is an urgent concern. This exclusion is not merely for permanent disabilities but extends also to temporary conditions (e.g., injury, pregnancy) and situational constraints (e.g., lack of space for a full-body shot), which makes technology unusable for a much larger segment of the population than is typically acknowledged.

2.2. The illusion of Representation and its Commercial Implications:

Alongside physical availability, there is a crisis of representation. The standard practice of generic, often Western-centered and light-skinned body models, combined with technical exclusion of appropriate skin tone analysis, renders virtual try-on experiences structurally unrepresentative for a global majority. The appearance of a garment relies heavily on how it sits atop the wearer's unique body shape and undertone of their skin.

When a system produces a dress on an average mannequin, it provides no valuable information to a user of a variable body type or darker skin tone. It erodes confidence in technology and, subsequently, the brands that employ it. Such a loss of confidence is not transitory; it fuels the fashion e-commerce sector's unsustainable return rates of 20-40% [1]. Every return is an absolute loss in packaging, logistics, and often the item itself, which can't be resold as new. This problem represents billions of dollars' worth of lost revenue and environmental damage annually, making it a leading commercial challenge.

2.3 Bias Spread and Ethical Problems:

There are many AI systems that mirror their training data. If body model algorithms are predominantly trained on underrepresented population data, their performance will always be suboptimal on underrepresented populations. This creates a cycle where technology fails diverse populations, thus deterring them from participating and again shifting future data capture towards unnatural demographics. The problem extends beyond body form to skin tones. Most computer vision algorithms are especially worst at consistently detecting and processing darker skin tones due to photographic film and image-processing algorithm historical biases [9]. By failing to actively address this, VTO systems can actually risk embedding and amplifying such biases by automating them, offering a subpar and often alienating experience to users of color and embedding discriminatory practices even further into the fashion business.

2.4 Economic and Scalability Barriers to Innovation:

The cost and operational sophistication associated with professional-grade 3D scanning technology (equipment, calibration, skilled operators) excludes widespread adoption by small- to medium-sized enterprises (SMEs) and incorporation into high-volume, mass-market e-commerce web sites. This is left for high-end luxury products or niche applications, which lower the technology's availability in general and its aggregate potential to bring innovation. It creates a market in which it is only the large players that have the resources to tackle the "digital fit" problem, thereby widening the competitive divide and deterring innovation in the broader retail landscape. The industry needs a democratic solution that is technically viable and economically pragmatic for firms of

all shapes and sizes.

Thus, there is an urgent and clear demand for paradigm shift. The answer must be a robust, software-solution that leverages the most recent advances in software solution machine learning to bridge these gaps. It must be:

- **Accessibility-First:** From a minimal, accessible 2D image from a regular smartphone camera, accepting varied postures.
- **Representation-Aware:** Actively incorporating accurate skin tone analysis and body shape diversity as non-negotiable, critical elements.
- **Ethically Crafted:** Designed with a specific effort to identify and counteract bias at each component of the AI pipeline.
- **Economically Scalable:** Offered as a cost-effective API or service that enables any retailer to offer a high-fidelity virtual try-on feature.

This research tackles this complex problem directly through the proposal of a system not just for geometric accuracy, but for human-centered inclusivity with the goal of a more equitable, more reliable, and more effective online shopping experience for all individuals.



Figure 1: Post estimation on varied postures

3 OBJECTIVES

3.1 Main Objective

The primary aim of this work is to pioneer the creation of a new, end-to-end AI-driven system that drastically brings access to high-fidelity virtual try-on technology. This will be achieved by creating and building a high-level software module that is able to smartly deduce a user's unique physiological features i.e., their precise body shape morphology and natural skin tone based upon a single, restricted, two-dimensional RGB picture, even one captured in suboptimal, real-world conditions (e.g., non-standard sitting postures, distracting backgrounds, inconstant lighting).

The module will take a synergistic blend of the most effective computer vision techniques for robust human pose estimation and semantic segmentation, advanced machine learning algorithms for anthropometric ratio computation and body typology classification, and a precisely designed pipeline of skin tone analysis using perceptually uniform color space transformation to obtain lighting-invariant accuracy. The peak of this multi-aspect analysis shall not be a simple statistical prediction, but the fully automated, parameter-driven construction of an individualized, detailed, and anatomically correct 3D virtual mannequin.

This virtual avatar shall be rigged and rendered in order to be utilized as the foundation, accurate, and dynamic model of the user within the greater Style Mate system. The ultimate long-term objective is to address head-on the extensive product return rate and customer dissatisfaction associated with internet fashion e-commerce by possessing a solution that is platform-independent and transcends dependency on expensive hardware, hence providing access to the type of inclusive, self-assured, and tailored virtual try-on experience absent in the internet fashion e-commerce industry. This research seeks to set a new standard for digital fashion accessibility and realism, from technical innovation to socio-economic impact.

3.2 Specific Objectives

3.2.1 Data extraction, pre-processing, cleaning, and structuring

This objective is focused on building a robust pipeline for obtaining and processing high-quality input data, the basis of any reliable AI system.

Task1: Develop a secure, user-friendly web interface module (integratable into the main StyleMate frontend) for simple image acquisition either by direct webcam input or upload, cross-browser compatible.

Task 2: Create an effective pre-processing pipeline that removes input data impurities. This will include:

- Advanced background elimination/cluttering through a pre-trained model like U²-Net to isolate the subject.
- Image normalization and contrast-limited adaptive histogram equalization (CLAHE) to eliminate lighting variations.
- Standardization of resolution and noise reduction methods to provide a uniform quality input for the subsequent models.

Task 3: Create a varied, ethically sourced training dataset. This will involve utilizing existing public datasets (SSP-3D, FairFace) and conducting limited-scale, IRB-approved primary data collection from a representative sample of volunteers to enhance model robustness for non-standard inputs and fairness in body types and complexion.

3.2.2 Body Landmark Detection and Anthropometric Shape Classification

This objective is dedicated to transforming a 2D image into accurate body shape information.

Task 1: Optimize and develop an innovative human pose estimation system (Google's MediaPipe Holistic) to correctly obtain a rich list of 2D body landmarks (33 pose, 21 hand, 468 face landmarks) from the pre-processed image, even under seated or partial views.

Task 2: Develop algorithms to calculate key anthropometric ratios (e.g., shoulder-to-hip ratio, waist-to-hip ratio, torso-to-leg ratio) and estimate proxy measurements using the 2D landmarks and correcting for perspective distortion.

Task 3: Train, validate, and test a machine learning classifier (for example, a Support Vector Machine or a Neural Network) for classifying the body shape into a generic typology (for example, Hourglass, Pear, Rectangle, Inverted Triangle, Oval) based on the calculated ratios and proportions.

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3.2.3 Adaptive and Lighting-Invariant Skin Tone Detection and Analysis

This objective ensures the virtual representation to be culturally sensitive and realistic.

Task 1: Develop an enhanced algorithm to precisely detect and segment skin regions from the facial and: Develop posed parts of the image using a combination of face detection algorithms and color thresholding in the YCbCr color space in an effort to minimize false positives.

Task 2: Apply color space transformations (RGB to HSV and CIELAB) on segmented skin pixels to ensure perceptual uniformity and lighting invariance to make the analysis invariant to photo conditions.

Task 3: Run a trained machine learning classifier (e.g., a Convolutional Neural Network or an instance like K-Means clustering) to classify the user's main skin tone according to an established dermatological scale (e.g., Fitzpatrick scale categories or the Monk Skin Tone Scale), reporting both a label and representative color values for texture application use.

3.2.4 3D Virtual Mannequin Parameterization and Dynamic Generation

This objective is tasked with producing the final 3D output from the extracted parameters.

Task 1: Develop an integration pipeline that smoothly maps Objectives 3.2.2 and 3.2.3 outputs (body shape category, anthropometric ratios, skin tone category/values) onto a parametric 3D human model parameters (e.g., modifying the SMPL model or employing the MakeHuman API).

Task 2: Develop scripts to programmatically interact with 3D modeling packages (Blender through its Python API 'bpy') to dynamically generate a personalized 3D mesh, apply skin texture and shaders relevant to the tone identified, and rig the model for potential animation.

Task 3: Enhance the output 3D model for deployment on the web so it's light, well-formatted (e.g., glTF/GLB format), and ready to be compiled with the rendering engine of the main StyleMate application.

3.2.5 Adaptive and Lighting-Invariant Skin Tone Detection and Analysis

This objective ensures the component will fit seamlessly into the overall system and meet quality standards.

Task 1: Package the whole pipeline as a containerized microservice (using Docker) with a well-specified RESTful API (coded using Flask/FastAPI) for seamless integration with the greater StyleMate framework, allowing seamless data transfer with the outfit recommendation and chatbot modules.

Task 2: Conduct a thorough, multi-faceted evaluation of the performance of the system. This includes:

Quantitative Analysis: Measuring the accuracy (Precision, Recall, F1-Score) of the body shape and skin tone classifiers on a test set withheld from training.

Performance Benchmarking: Measuring end-to-end inference time from submission of image to generation of model.

Qualitative User Studies: UAT with participants of various backgrounds to gather feedback about perceived realism, accuracy, and satisfaction with the generated avatar using a Mean Opinion Score (MOS) and System Usability Scale (SUS).

4 METHODOLOGY

Methodology of the proposed system, StyleMate, aims to overcome limitations achieved in the research gap as well as deliver an accessible, inclusive, and scalable virtual try-on system. The architecture is multi-layered, integrating computer vision, machine learning, and human–computer interaction principles in a unified pipeline. As opposed to other research that either utilizes highly controlled input conditions or costly 3D hardware setups, the system in this method is made to operate with general 2D inputs such as smartphone or webcam photos. Through accommodation for seated users such as the depiction of skin color and inclusion of accessibility from the outset, the proposed methodology ensures user inclusivity on a wide variety of potential users.

The following subsections describe the system architecture, mannequin development process, recommendation system, accessibility interaction layer, tools and technologies, and evaluation strategy for StyleMate. Each subsection is described elaborately to demonstrate how it works cooperatively to accomplish the objectives of this research.

4.1 System Architecture diagrams

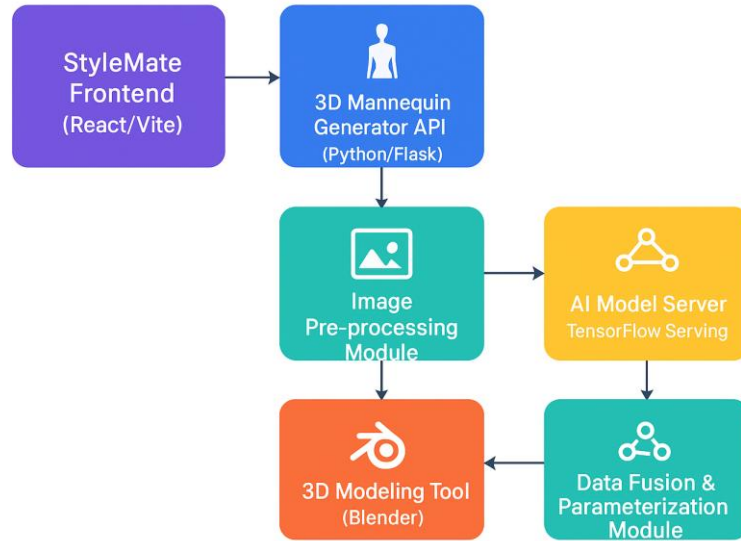


Figure 2: High-level System Architecture

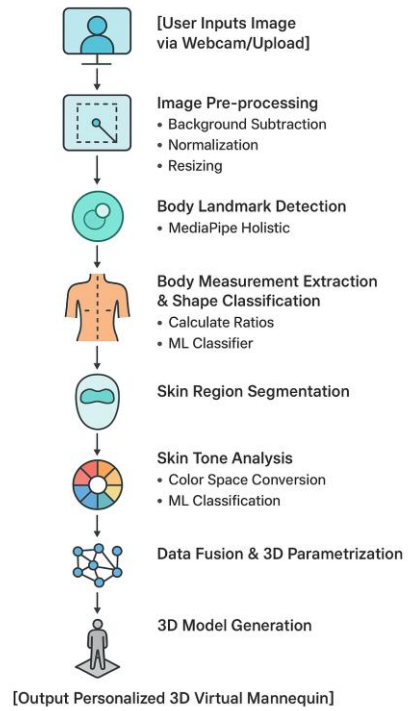


Figure 3: Component-Specific Process Flow

4.2 Data Collection

A multi-view data acquisition plan will be employed to ensure model robustness and fairness.

- **Public Datasets:** Public datasets will be used for initial training.
- **Body Shape:** The SSP-3D dataset [8] or similar, which contains 3D scans with associated shape parameters, will be employed for training the body shape inference model.
- **Skin Tone:** The FairFace dataset [7] provides a large, balanced dataset of facial images with race, age, and gender labels, ideal for training a stable and fair skin tone classifier.
- **Synthetic Data Generation:** The existing data will be augmented through operations like rotation, scaling, and brightness adjustment to improve model generalization under varying lighting conditions and angles.

Ethical Small-Scale Custom Data Collection: Small-scale, IRB-approved photography will be conducted by a diverse group of volunteers (with permission).

Photo-shoots in various poses (sit, stand) and lighting will be done to improve the performance of the model on "in-the-wild" data specifically and offer inclusivity.

4.3 Software Solution

The software development methodology which would be followed throughout the development of this system would be the Agile and SCRUM methodologies. This would enable the iterative and flexible development of the system and enable teams to adapt to changing requirements throughout the development process.

Stakeholder feedback is given a lot of weight in the agile technique, along with teamwork and communication. It comprises breaking up the development cycle into brief, iterative "sprints". Another way the agile methodology encourages continual improvement is through regular retrospectives, in which team members assess their performance and search for opportunities for improvement [24].

Scrum is a specific framework for applying agile principles, with a focus on project management and team composition. Regular meetings will be held under Scrum, including daily stand-ups to talk about progress and get input from stakeholders [25].

In conclusion, Agile and Scrum approaches would be applied during the system's development process to make sure that it is created iteratively and collaboratively with frequent chances for input and course correction. With an emphasis on creating a high-quality product that meets the needs of its customers, these strategies can help guarantee that the system is developed in a timely and effective manner.



Figure 7: Agile Methodology

Source: project-management.com (January 29, 2023)

4.4 Requirement Gathering and Analysis

4.4.1 Functional requirements

The functional requirements derived from this exercise are what the system must do to meet user needs and address the research gap. Essentially, the system must allow the user to upload or capture an image using a basic camera or webcam. The image must be pre-processed, with that including background subtraction, normalization, and pose estimation. Following this, the system should detect prominent landmarks, calculate measurements, and construct mannequins to mirror body form and complexion in the same way. Beyond the construction of mannequins, the system should place selected clothing in a realistic fashion, providing users with a legitimate preview of fit and style.

In addition to this, the functional scope also includes a mood-sensitive recommendation engine that provides clothing suggestions based on the moods and surroundings of the users, providing accurate and relevant suggestions for the user. To ensure accessibility, the system also must be capable of supporting alternative means of interaction such as voice commands, text-to-speech, and low-touch procedures, so differently abled users can utilize the platform without any problems. These requirements define the functional nature of StyleMate and ensure that it performs on the deficits observed in existing systems.

4.4.2 Non-functional requirements

Beyond functional needs, the system also needs to satisfy non-functional needs defining its performance, quality, and reliability. The system must initially be responsive, as it takes just a couple of seconds to produce mannequins and outfit previews to avoid irritating the user. Scalability is also important because the platform must support various users simultaneously if it is connected to online retail sites. Reliability and availability are also necessary, as the system must remain operational even under constant use.

Usability forms another corner of the non-functional requirements. Global accessibility standards such as WCAG 2.1 must be adopted by the platform in order for the design to be accessible to individuals with varying abilities. Security and privacy are also not negotiable properties, as the users will be uploading their individual images. Storage and transfer of data must therefore be encrypted, and secure transmission mechanisms

must be available. Finally, the compatibility with browsers and devices ensures that the system is not restricted to specific environments but can be used on a large scale.

4.4.3 Feasibility study

The feasibility of StyleMate was assessed from three key viewpoints: technical, operational, and economic. From a technical viewpoint, the proposed system is highly feasible since it is built upon established and well-supported technologies such as MediaPipe BlazePose for pose estimation, GANs for overlaying clothes, and CNNs for body shape classification. These technologies are widely available as open-source frameworks and can be deployed on standard cloud infrastructure, which is a testament to the deployment feasibility.

Operational feasibility was checked based on user adoption and ease of integration. The requirements gathering process confirmed that the users, particularly from Sri Lanka and other developing nations, require low-cost systems with easy usage that run using readily available devices. The proposed solution does not employ expensive devices like 3D scanners but exploits readily available smartphone or webcam images. This assists in ensuring that the site is not only operational but also attractive to retailers looking for scalable solutions with wide appeal to varied customer bases.

Finally, the economic feasibility study indicated that the system can be constructed at reasonable cost since it is developed on open-source platforms and elastic cloud infrastructure. Furthermore, the solution is highly likely to recover costs and grow through integration with web commerce programs, in which online shops can achieve increased customer satisfaction and reduced return rates. Taken together, these findings confirm that StyleMate is technologically feasible, operational feasibility, and economically feasible.

4.5 Tools and Technologies

The development of the system requires an extensive amount of Information Technology related tools and technologies spanning from Integrated Development Environments (IDE) to development languages.

4.5.1 Tools

- **Visual Studio Code** - Visual Studio Code is a quick yet effective source code editor that runs on Windows, macOS, and Linux. It contains a robust ecosystem of extensions for additional languages and runtimes (including C++, C#, Java, Python, PHP, Go, and .NET). Visual Studio Code has been popular among developers due to its ease of use, versatility, and cross-platform compatibility. Visual Studio Code is a helpful tool for developers who wish to streamline their workflow and boost productivity because of the support from the community.
- **GitLab** – GitLab is an online Git repository manager in addition to source code management, continuous integration and deployment, and other DevOps tools. It offers a complete solution for managing the software development lifecycle, from source code management to code review, testing, and deployment. GitLab provides a web-based interface for managing repositories and carrying out code reviews.
- **Google Colab** – Without the requirement for setup or installation, Google Colab (short for Google Colaboratory) enables users to create and run Python apps in a web browser. It gives users access to advanced computational resources including CPUs, GPUs, and TPUs, and is based on Jupyter notebooks. Users of Google Colab can execute machine learning tasks using well-known libraries like TensorFlow, PyTorch, and Scikit-learn, show data, build and run Python code, and more.
- **Trello** – Trello is an online project management tool that allows individuals and groups to plan and manage tasks, projects, and workflows. Users can track and visualize their work progress by utilizing a set of lists, cards, and boards.

- **Microsoft Teams** – Teams is an online communication platform developed by Microsoft. Teams platform is utilized to maintain communication between the development team and to hold meetings with stakeholders it has also been used for document management and has been linked to OneDrive.

4.5.2 Technologies

- **Python** - Python is a high-level programming language that is widely used by developers due to its ease of use and versatility. Python's emphasis on code readability and simplicity is one of its defining characteristics. It has syntax similar to that of natural language. It has numerous applications, including web development, scientific computing, data analysis, artificial intelligence, and others.
- **Scikit-learn** - Scikit-learn is a library that includes algorithms for both supervised and unsupervised learning. It is based on NumPy, SciPy, and matplotlib, making integration with other Python applications simple.
- **TensorFlow** - TensorFlow is an open-source software library developed by Google. This is adaptable and scalable. TensorFlow is appropriate for a wide range of applications from research to production deployment and it provides a simple and intuitive modeling interface.
- **PyTorch** - PyTorch is a free and open-source deep learning machine learning library. It has a dynamic computational graph that enables quick model creation and modification.
- **Libraries for Natural Language Processing (NLP)** - These libraries would be required to analyze and process large amounts of data. In NLP tasks, libraries such as NLTK, spaCy, and CoreNLP are widely used.

4.6 Implementation

Implementation will be into sequential, manageable steps.

- Phase 1: Foundation & Pre-processing: Set up the development environment. Install the image capture/upload module and the pre-processing pipeline (background removal using U²-Net, normalization).

- Phase 2: Core CV & ML Models:

Implement and validate pose estimation and landmark extraction with MediaPipe.

Implement the computation logic for body ratios and train the body shape classifier (e.g., SVM, Random Forest) on SSP-3D. Implement the skin tone classification and skin segmentation model based on the FairFace dataset.

- Phase 3: 3D Integration: Implement Python scripts to control Blender. Create functions to build a basic 3D model with input parameters (shape, height). Implement texture mapping for the application of skin tone.
- Phase 4: API Development & Pipeline Integration: Integrate all modules into a single Python pipeline. Distribute the pipeline as a Flask API with an image input and an output of a 3D model.
- Phase 5: Deployment, Refinement & Testing: Conduct unit testing, integration testing, and user acceptance testing (UAT). Improve models based on user reviews. Containerize the application using Docker for cloud platform deployment.

4.7 Deployment

- **Google Cloud Platform (GCP)** - The use of GCP to host the Sri Lankan legal information retrieval system provides a dependable and scalable solution for hosting the application and managing its data. GCP provides efficient and cost-effective infrastructure for deploying and managing applications. GCP also provides monitoring and management tools for deploying applications that

enable real-time monitoring and analysis of application performance, assisting in the rapid identification and resolution of issues.

4.8 Evaluation of the System

The system's success will be measured with a combination of technical and user-centric measures:

- Technical Accuracy:
 - Body Shape Classification: Accuracy, Recall, F1-Score against a human-annotated test set.
 - Skin Tone Classification: Accuracy and equity across diverse skin tone classes.
 - Landmark Detection Accuracy: Mean Average Precision (mAP) across detected key points.
- Performance:
 - Inference Time: Total processing time from image input to 3D model output (Target: < 60 seconds for a proof-of-concept).
- User Satisfaction:
 - Mean Opinion Score (MOS): The respondents give a score of the accuracy and realism of their designed avatar on a 1 (Very Poor) to 5 (Excellent) scale.
 - System Usability Scale (SUS): A standardized questionnaire for assessing respondents' self-reported ease of use of the image upload procedure purpose.

5 WORK BREAKDOWN STRUCTURE

A Work Breakdown Structure (WBS) is a project's hierarchical division into smaller, more manageable components or work packages. It is a graphical representation of the project scope that divides it into deliverables and sub-deliverables. The work breakdown structure (WBS) enables effective task organization and planning, ensuring that all project components are accounted for and completed on time. The structure can be presented graphically or tabularly, as shown below:

According to the above WBS, the project would be carried out sequentially. This would streamline the planning, documentation, implementation, integration, validation, and deployment of the system and would produce clear and concise tasks for the developers to follow.

6 GANTT CHART

The Gantt chart below shows and tracks the progress of the project's tasks and activities over time. This visual representation of the project timeline aids the team's understanding of the overall project plan and the identification of critical tasks or milestones.

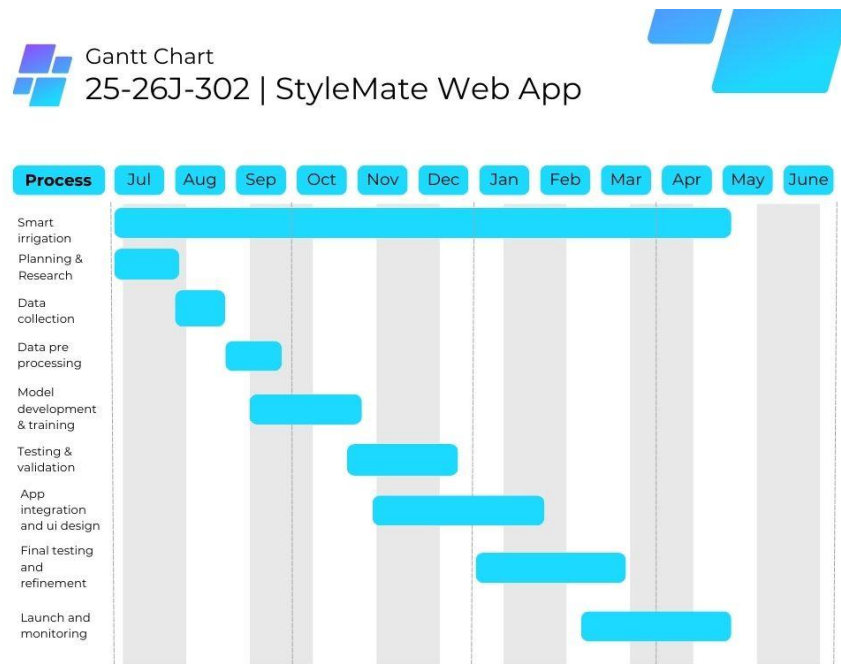


Figure 9: Gantt chart

7 COMMERCIALIZATION OF THE PRODUCT

The projected 3D Virtual Mannequin Generator is not merely a technological innovation; it is a compelling business proposition with a clearly articulated value hypothesis. Its core objective to enhance consumer trust and accuracy when buying clothes online significantly hits at the biggest problem in global fashion e-commerce: product returns. The following section outlines a commercialization plan with focus on market discovery, competitive advantage, and scalable business models and augmented with empirics whenever appropriate.

7.1 Target Market

With The solution is scalable for adoption across different market verticals of the digital retail world. The target market is segmented into three important tiers, each with a different use case and value proposition.

Primary Market Segments:

Large Enterprise E-Commerce Platforms: This segment includes global online players (e.g., Amazon, Zalando) and large fashion brands with big digital sales channels. The most important pain area is the enormous financial cost of loss owing to return rates, up to 30% for apparel, totaling billions of dollars in logistics, re-stocking, and lost sales annually [1]. The value proposition is basically a reduction of these return rates directly through providing a greater fit and style preview, which translates into tremendous cost savings and increased customer lifetime value.

Small and Medium-sized Businesses (SMEs) & Direct-to-Consumer (D2C) Brands: This segment consists of new, emerging brands and established SMEs selling predominantly through direct-to-consumer platforms. Their pain point is to compete against large establishments and high customer acquisition costs. The value proposition here is competitive differentiation. Introducing innovative virtual try-on technology can enhance conversion rates, AOV, and brand loyalty through an innovative and customer-centric shopping experience [14].

Business-to-Business (B2B) Software-as-a-Service (SaaS) Providers: These include providers of e-commerce platforms (e.g., Shopify add-ons), augmented reality (AR) products, or fashion technology APIs. Their need is to introduce new high-value features on a regular basis to their product lines. The value proposition is a licensable, API-based "as-a-service" component that can be white-labeled and appended to their existing offerings, creating new revenue for them and for this initiative in the form of licensing revenue.

Market Validation: The global fashion online retail market shall see growth on a grand scale, and this indicates the perfect time for enabling technologies. The urgent need for answers regarding returns is a good entry point for this technology [15].

7.2 Monetization Strategy

In A multi-level monetization strategy is suggested to cater to the different segments found and to attain optimum market penetration.

Pay-Per-Use API Model: This would involve imposing a charge for every successful 3D model generated via an API call. It is best applied for Tier 1 (large platforms) and Tier 3 (SaaS providers) customers as it directly links costs to their level of usage, without a large up-front commitment and with built-in scalability.

Tiered Subscription Model: Monthly or annual subscription plans can be offered based on the anticipated level of usage (e.g., number of generations, active users). This is a forecastable costing for Tier 2 clients (SMEs and D2C brands) and a standard, familiar practice in SaaS, lowering the barrier to adoption.

White-Label Licensing: For known technology providers (Tier 3), premium white-label licensing can be offered. This would be an upfront licensing fee in exchange for the rights to fully integrate the technology under their own brand, integrating the solution into their product suites and using their established distribution channels.

The particular pricing parameters for the models would be determined by a detailed market research, competitor benchmarking with prioritized focus, and customer discovery interviews post-prototype validation, which falls outside the scope of this academic exercise currently.

7.3 Path to Market and Initial Strategy

The path to market to start with will be the most effective channel for adoption:

- **Primary Focus:** Initially target B2B SaaS providers (Tier 3). This is operationally viable, in the sense that a single integration partnership can directly expose the technology to the entire existing client base of the provider, eliminating the need for one-at-a-time B2C sales cycles to initiate.
- **Secondary Focus:** Simultaneously pilot with a small set of Direct-to-Consumer brands (Tier 2). This would be utilized to develop validated case studies, testimonials, and hard metrics on KPIs like conversion rate lift and return rate reduction that are critical for future sales and marketing collateral.
- **Content Marketing:** To build credibility, findings and outcomes of the pilot program can be summarized in the form of whitepapers and case studies and reach decision-makers (e.g., VPs of E-commerce) through industry publications and professional networks such as LinkedIn.

8 BUDGET AND BUDGET JUSTIFICATION

Category	Description	Estimated Cost (LKR)
Hosting & Cloud	Basic cloud server / GPU rental for 6 months	30,000 .00
Domain & Web Hosting	Domain (stylemate.lk) + basic web hosting	10,000.00
Publication	Conference/journal fees + printing/binding	12,500 .00
Deployment	Cloud deployment of prototype (3–4 months)	15,000.00
Marketing (Optional)	Posters, brochures, small-scale promotions	20,000.00
Total Estimated		87,500.00

Table 2: Budget Justification

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